

Open Modular Scattering Under Trace Closure: Endpoint Coboundaries, Selberg Loops, and Commuting Squarefree Cusp Channels

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Abstract

We test whether modular cusp scattering retains genuinely open arithmetic when one closes it by an ordinary trace or determinant. The test changes the mathematical object in two ways left open by a previous final-denominator obstruction: it retains boundary endpoints, and it retains every cusp channel of squarefree congruence surfaces. First, unoriented open scattering geodesics have an explicit Dirichlet–Laplace series containing both a totient zeta quotient and a χ_{-4} term; this provides a non-Selberg positive control. On the rational endpoint action groupoid, however, every section-induced absolute projective automorphy cocycle of the specified form is an algebraic coboundary. The affine value $2 \log |c|$ is exactly an endpoint denominator-gauge difference. Enlarging the boundary creates loops with nonzero automorphy period, but those periods are precisely the signed hyperbolic translation lengths. Independently, the complete standard scattering family for squarefree $\Gamma_0(N)$ has a fixed Walsh–Hadamard eigenbasis. We preserve and multiply the matrices in a frozen order, then prove that the product is permutation-invariant because the source algebra is commutative. The spectral parameter is not asserted to be dynamical time. Endpoint projectors restore parameter-to-edge assignment and path sensitivity, delimiting the obstruction. Exact arithmetic through denominator 2000 and 80-digit matrix checks through the 16-cusp level $N = 210$ audit the formulas without zero tables or fitted parameters. These results reject the frozen rational ordinary-loop construction and conditional bare-product proposal, not off-diagonal kernels or separately derived operators.

1 Introduction

The arithmetic in cusp scattering is naturally open: a trajectory enters from infinity, crosses a compact core, and leaves toward an endpoint. A dynamical zeta or Fredholm determinant is naturally closed: its trace expansion identifies the final state with the initial state and organizes repetitions of primitive loops. The distinction matters in Hilbert–Pólya searches. A completed zeta function can occur in a scattering coefficient without the scattering channels providing the primitive periods or discrete self-adjoint spectrum required by the proposed construction.

The deterministic area-preserving Hénon program that motivates this project asks for a source-derived route from nonlinear dynamics to an arithmetic spectral object [14]. Earlier tests showed that the final denominator of a modular matrix gives exact open-channel arithmetic but cannot serve as a nonzero homogeneous clock on closed hyperbolic classes. That result did not address two large repairs: retain the endpoint so that denominator increments compose locally, or replace the one-cusp scalar by a multi-cusp matrix and retain all channels. We test both repairs here without assuming that either is a dynamical zeta.

Scattering geodesics and their sojourn times are classical [4, 6, 7]; a recent modular count gives particularly explicit endpoint data [12]. We begin with that open object rather than forcing

it into a closed-orbit format. Its unoriented Laplace series is

$$Z_{sc}(s; T_0) = \frac{T_0^{-2s}}{2} \left[\frac{\zeta(2s-1)}{\zeta(2s)} + \frac{\zeta(2s)L(2s, \chi_{-4})}{\zeta(4s)} \right]. \quad (1)$$

This formula is a direct corollary of the known count and is not presented as a new scattering theory. Its purpose is sharper: open arithmetic survives before diagonal closure.

Our first main result identifies what endpoint retention actually changes. The double-coset space $P \backslash \mathrm{PSL}_2(\mathbb{Z})/P$ does not compose representative-independently. The boundary action groupoid does. A projective section defines an exact automorphy cocycle, but on $\mathbb{P}^1(\mathbb{Q})$ the primitive integral section makes the cocycle identically zero. Every other section differs by an algebraic coboundary. In particular, $2 \log |c|$ in the affine section is a difference of endpoint denominator gauges rather than an intrinsic loop period.

The second main result describes the cost of creating nonzero trace periods. On the full real boundary, nonidentity loops are parabolic rational fixed points or hyperbolic quadratic fixed points. The former have zero automorphy period. At the attracting and repelling hyperbolic fixed points the periods are respectively plus and minus the translation length. Therefore the automorphy-period support of a standard loop trace is zero on the rational cusp orbit and otherwise lies in the signed Selberg length spectrum. This conclusion agrees with the established modular transfer-operator endpoint [9, 11], but the elementary groupoid dichotomy makes the open-to-closed loss explicit.

The multi-cusp test reaches the same boundary by a different mechanism. The standard scattering matrix of squarefree $\Gamma_0(N)$ has the classical tensor form of Huxley and Hejhal [5, 2, 8]. A fixed Walsh transform diagonalizes every spectral parameter, producing only the modular completed-zeta quotient times finite Euler factors. Consequently, the source matrices commute. We do not average their entries: we form a frozen product at distinct spectral parameters first and prove that it is invariant under reordering. This is a diagnostic for a proposal that elects to encode successive steps by such bare products; scattering theory itself does not identify the spectral parameter with time.

The obstruction has a necessary boundary. Inserting rank-one cusp projectors between scattering matrices leaves the commutative scattering algebra. Such projector-resolved amplitudes can distinguish parameter-to-edge assignments and endpoint paths. Related off-diagonal and twisted zeta functions already have a substantial literature [13, 3]; a finite witness alone does not define another Fredholm determinant.

The contributions are therefore the following.

1. We derive equation (1) as a genuinely open positive control and audit its coefficients exactly through $q = 2000$.
2. We prove that the rational-endpoint automorphy cocycle is an algebraic coboundary, while its nonzero full-boundary loop periods are exactly signed Selberg lengths.
3. We prove simultaneous Walsh diagonalization and permutation invariance for frozen spectral-parameter products in the standard squarefree scattering algebra, verifying the identities through $N = 210$ at 80-digit precision.
4. We exhibit projector-resolved assignment and path sensitivity and state it as an explicit exclusion from the no-go rather than suppressing the positive signal.

All headline claims are analytic. Computation audits normalization, matrix identities, and implementation; no claim is fitted to Riemann-zero data. The result is a scoped Route-A obstruction, not a proof of the Riemann hypothesis or a universal theorem about open groupoid kernels.

2 A genuinely open arithmetic series

Fix a sufficiently large compact-core height $T_0 > 1$. Pujahari and Satpathy classify unoriented modular scattering geodesics by subsets of reduced rationals [12]. If s_q denotes the number of solutions to $r^2 \equiv -1 \pmod{q}$, with $s_1 = 1$, their result gives

$$n_q = \frac{\varphi(q) + s_q}{2}, \quad \ell_q = 2 \log(qT_0). \quad (2)$$

The special term $q = 1$ represents the rational endpoint zero.

Proposition 2.1 (open scattering series). *For $\Re s > 1$, the Laplace series over the unoriented scattering geodesics is*

$$Z_{\text{sc}}(s; T_0) = \frac{T_0^{-2s}}{2} \left[\frac{\zeta(2s-1)}{\zeta(2s)} + \frac{\zeta(2s)L(2s, \chi_{-4})}{\zeta(4s)} \right]. \quad (3)$$

It has meromorphic continuation through the right-hand side, with

$$\text{Res}_{s=1} Z_{\text{sc}} = \frac{3}{2\pi^2 T_0^2}, \quad \text{Res}_{s=1/2} Z_{\text{sc}} = \frac{3}{8\pi T_0}. \quad (4)$$

Proof. The standard totient identity gives

$$\sum_{q \geq 1} \frac{\varphi(q)}{q^w} = \frac{\zeta(w-1)}{\zeta(w)}.$$

For the second coefficient, roots of -1 lift uniquely through powers of a prime congruent to one modulo four, do not exist at primes congruent to three modulo four, and contribute the factor $1 + 2^{-w}$ at two. Thus

$$\sum_{q \geq 1} \frac{s_q}{q^w} = (1 + 2^{-w}) \prod_{p \equiv 1(4)} \frac{1 + p^{-w}}{1 - p^{-w}} = \frac{\zeta(w)L(w, \chi_{-4})}{\zeta(2w)}.$$

Substitute $w = 2s$ into equation (2). The residues follow from $L(1, \chi_{-4}) = \pi/4$ and $\zeta(2) = \pi^2/6$. $\square$

The object in equation (3) is a sum over open geodesics. It has no primitive-power law analogous to a closed-orbit Euler product. That is not a defect of the formula; it is the positive control against which trace closure will be measured. The coefficient audit later sets $T_0 = 1$ only as an analytic normalization of this Dirichlet series; that boundary value is not presented as a positive geometric sojourn-time cutoff.

3 Endpoint retention and rational gauge triviality

Let P be the stabilizer of infinity in $\text{PSL}_2(\mathbb{Z})$. The open channels are naturally indexed by $P \backslash \text{PSL}_2(\mathbb{Z}) / P$, but a double-coset set is not automatically a category of composable paths.

Proposition 3.1 (double-coset composition is not intrinsic). *Multiplication of representatives does not define a product on $P \backslash \text{PSL}_2(\mathbb{Z}) / P$.*

Proof. With

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

the matrices S and ST^n represent the same right P -coset. Yet

$$SS = -I, \quad ST^n S = \begin{pmatrix} -1 & 0 \\ n & -1 \end{pmatrix}.$$

For $n \neq 0$, the first product is in the identity cell and the second has nonzero lower-left entry, so they lie in different double cosets. $\square$

The endpoint action groupoid repairs this type error. For a $\mathrm{PSL}_2(\mathbb{Z})$ -invariant subset $X \subseteq \mathbb{P}^1(\mathbb{R})$, an arrow (g, x) runs from x to gx , and $(g, hx) \circ (h, x) = (gh, x)$. Choose a nonzero vector v_x on each projective line. Define

$$gv_x = j_v(g, x)v_{gx}, \quad \sigma_v(g, x) = 2 \log |j_v(g, x)|. \quad (5)$$

Then

$$\sigma_v(gh, x) = \sigma_v(g, hx) + \sigma_v(h, x). \quad (6)$$

This is the elementary projective form of a Busemann-type groupoid cocycle; the general framework is developed in [10].

Theorem 3.2 (rational cusp algebraic coboundary). *Every section-induced absolute projective cocycle of the form equation (5) on $\mathrm{PSL}_2(\mathbb{Z}) \ltimes \mathbb{P}^1(\mathbb{Q})$ is a set-theoretic groupoid coboundary. In the affine section,*

$$\sigma_{\mathrm{aff}}(g, x) = 2 \log q(gx) - 2 \log q(x), \quad (7)$$

where $q(p/q) = q$ in lowest terms and $q(\infty) = 1$. In particular, if $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $c \neq 0$,

$$\sigma_{\mathrm{aff}}(g, \infty) = 2 \log |c|.$$

Proof. For a reduced rational $x = p/q$, choose a primitive vector $\hat{v}_x = (p, q)^T$, and take $\hat{v}_\infty = (1, 0)^T$. An integral unimodular matrix maps a primitive integral vector to another primitive integral vector. After the fixed sign convention is restored,

$$g\hat{v}_x = \pm \hat{v}_{gx}.$$

Hence the associated absolute automorphy factor is one and $\sigma_{\hat{v}} \equiv 0$.

Every other projective section has the form $v'_x = r(x)\hat{v}_x$. Its cocycle differs from zero by

$$2 \log |r(x)| - 2 \log |r(gx)|.$$

For the affine section, $r(x) = q(x)^{-1}$, which proves equation (7) and the specialization at infinity. $\square$

Thus endpoint retention makes the denominator height exactly additive, but only as an endpoint gauge difference. It creates no nonzero intrinsic loop period on the rational cusp orbit.

The qualifier “set-theoretic” is essential. The primitive section and the denominator gauge are neither continuous nor bounded for the topology inherited from $\mathbb{P}^1(\mathbb{R})$. The theorem does not assert a continuous, Hölder, or Livšic coboundary and supplies no bounded conjugacy of analytic transfer operators. Its trace-level consequence is only that this coboundary has zero period on rational loops.

There is also a parity obstruction. The inverse of (g, x) is (g^{-1}, gx) , so every real cocycle obeys

$$\sigma(g^{-1}, gx) = -\sigma(g, x).$$

A physical sojourn time is positive and unchanged by reversing the geodesic. It therefore cannot itself be a nonzero real groupoid cocycle; additivity requires a signed lift.

Table 1: The same modular data at three stages. Open arithmetic survives off diagonal; the two ordinary loop closures are either period-trivial or Selberg-periodic.

state space	primitive data	nonzero period	analytic object
open channels	cuspidal geodesics	sojourn time	Dirichlet series
rational endpoints	composable arrows	none intrinsic	coboundary
full real boundary	action-groupoid loops	$\pm\ell(g)$	classical Selberg length data

4 Full-boundary loops are Selberg periods

The rational cusp orbit is too small to carry nonzero automorphy periods. One might therefore enlarge the unit space to $\mathbb{P}^1(\mathbb{R})$ before taking a trace. The resulting loops are completely rigid.

Theorem 4.1 (cusp-trivial/full-boundary-Selberg dichotomy). *Let (g, x) be a nonidentity loop in $\mathrm{PSL}_2(\mathbb{Z}) \ltimes \mathbb{P}^1(\mathbb{R})$. If x is rational, then g is parabolic and $\sigma_v(g, x) = 0$. Otherwise x is a real quadratic irrational, g is hyperbolic, and its two fixed points satisfy*

$$\sigma_v(g, x_+) = \ell(g), \quad \sigma_v(g, x_-) = -\ell(g), \quad \ell(g) = 2 \operatorname{arccosh} \frac{|\operatorname{tr} g|}{2}. \quad (8)$$

The loop period is independent of the projective section.

Proof. A loop at $x = \infty$ has $c = 0$; after excluding the identity in $\mathrm{PSL}_2(\mathbb{Z})$, its stabilizer is parabolic and its automorphy period is zero. A finite fixed point satisfies

$$cx^2 + (d - a)x - b = 0$$

for an integral lift $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Nonidentity elliptic elements have no real fixed point. A parabolic element has one rational eigenline with eigenvalue of modulus one, hence zero automorphy period.

A hyperbolic element has two real eigenlines. Neither can be rational: a primitive integral eigenvector would give a rational algebraic-unit eigenvalue, hence ± 1 , contradicting hyperbolicity. The fixed points are therefore quadratic irrationals. If $\lambda > 1$ is the modulus of the expanding eigenvalue, the two automorphy factors have moduli λ and λ^{-1} . Their periods are $\pm 2 \log \lambda$, which equal the translation lengths in equation (8). A section change is a coboundary, and that coboundary vanishes on a loop. $\square$

Corollary 4.2 (period-support consequence). *Any standard trace expansion supported on loops of the frozen endpoint action groupoid has no nonzero rational-cusp period. After extension to the full real boundary, its nonzero automorphy-period set is the signed Selberg length spectrum.*

The word “standard” limits the corollary to a diagonal trace whose fixed points are action-groupoid loops. It does not cover off-diagonal matrix coefficients, extra fiber state, or inserted projectors. Within that scope, the period dichotomy is consistent with established nuclear modular transfer operators that close on Selberg zeta [9, 11]: nonzero trace periods arise only after the endpoints form a hyperbolic closed orbit. The dichotomy alone does not determine coding multiplicities, fixed-point Jacobians, nuclearity, or a Fredholm determinant.

5 Squarefree cusp channels form a commutative scattering algebra

Endpoint retention is not the only possible repair. For squarefree N , the surface $\Gamma_0(N) \backslash \mathbb{H}$ has $2^{\omega(N)}$ cusps. Its full standard scattering matrix retains every incoming and outgoing cusp label.

The classical computation, first obtained in the congruence scattering theory of Huxley and Hejhal and printed in modern sources [5, 2, 8], is

$$\Phi_N(s) = \phi(s) \bigotimes_{p|N} M_p(s), \quad \phi(s) = \frac{\Lambda(2s-1)}{\Lambda(2s)}, \quad (9)$$

where $\Lambda(w) = \pi^{-w/2} \Gamma(w/2) \zeta(w)$ and

$$M_p(s) = \frac{1}{p^{2s}-1} \begin{pmatrix} p-1 & p^s - p^{1-s} \\ p^s - p^{1-s} & p-1 \end{pmatrix}. \quad (10)$$

Young's explicit cusp/character basis changes place this factorization in the broader Eisenstein newform theory [15].

Theorem 5.1 (fixed Walsh diagonalization). *Let N be squarefree and $r = \omega(N)$. Fix the Huxley–Hejhal width/Atkin–Lehner cusp scaling matrices, order the divisor-labelled cusps by their prime-incidence bits, and use the tensor identification in equation (9). The matrix*

$$H_N = \bigotimes_{p|N} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

is independent of s and diagonalizes $\Phi_N(s)$ for every regular s . For $\varepsilon \in \{\pm 1\}^r$, the eigenchannel is

$$\phi_\varepsilon(s) = \phi(s) \prod_{p|N} \frac{1 + \varepsilon_p p^{1-s}}{1 + \varepsilon_p p^s}. \quad (11)$$

Furthermore,

$$\det \Phi_N(s) = \phi(s)^{2^r} \prod_{p|N} \left(\frac{1 - p^{2-2s}}{1 - p^{2s}} \right)^{2^{r-1}}. \quad (12)$$

Proof. Each local block is $a_p(s)I + b_p(s)X$, where X swaps the two local cusp bits. Its fixed eigenvectors are $(1, \pm 1)^T$, and direct factorization gives

$$a_p(s) + \varepsilon b_p(s) = \frac{1 + \varepsilon p^{1-s}}{1 + \varepsilon p^s}.$$

Tensoring proves equation (11). Multiplying the two local eigenvalues gives $(1 - p^{2-2s})/(1 - p^{2s})$. Each local determinant occurs 2^{r-1} times in the full tensor determinant, while $\phi(s)$ occurs in all 2^r channels, proving equation (12). $\square$

Corollary 5.2 (permutation invariance of spectral-parameter products). *For regular $s_1, \dots, s_m$ and every permutation π ,*

$$\Phi_N(s_m) \cdots \Phi_N(s_1) = \Phi_N(s_{\pi(m)}) \cdots \Phi_N(s_{\pi(1)}). \quad (13)$$

Proof. In the fixed Walsh basis the two products have diagonal entries $\prod_j \phi_\varepsilon(s_j)$. Scalar multiplication is permutation-invariant. $\square$

The proof preserves the frozen product and discovers that it is permutation-invariant. It does not replace the matrices by a mean transition matrix. The fixed basis change is also invertible and therefore does not by itself discard a single matrix. The obstruction is that the entire frozen family lies in one commutative Atkin–Lehner algebra. Here s remains a spectral parameter, not a source-derived evolution time. The result rules out chronology only conditionally: a model that encodes its successive steps by bare factors $\Phi_N(s_j)$ cannot recover their order from the

product. An s -dependent cusp renormalization is outside the fixed normalization and is not claimed to preserve simultaneous diagonalization.

The completed-zeta functional equation and the local ratios imply

$$\Phi_N(s)\Phi_N(1-s) = I.$$

On $\Re s = 1/2$, every ratio in equation (11) has modulus one, so the standard scattering matrix is unitary. These are strong analytic properties, but they do not create a primitive orbit or evolution law.

Finally, let ρ be a nontrivial zeta zero. A local numerator in equation (11) can vanish only on $\Re s = 1$, and a local denominator only on $\Re s = 0$. Thus no local factor cancels the pole of ϕ at $s = \rho/2$ or its zero at $s = (1 + \rho)/2$. Every standard eigenchannel retains the same shifted quotient divisor.

6 Endpoint projectors restore assignment and path sensitivity

The commutative-algebra result does not justify deleting cusp endpoints. Let $P_a = e_a e_a^T$ project onto one cusp in the standard basis. For three scattering steps,

$$\text{tr}(P_a \Phi(s_1) P_b \Phi(s_2) P_c \Phi(s_3)) = \Phi(s_1)_{ab} \Phi(s_2)_{bc} \Phi(s_3)_{ca}. \quad (14)$$

The projectors need not commute with the scattering matrices. Therefore equation (13) does not imply that equation (14) is invariant when parameters are reassigned to edges or when the endpoint path is changed.

Level $N = 6$ is the smallest squarefree level with two independent cusp bits and three distinct nontrivial flip types. Its four cusps can be labelled by the two bits associated with $p = 2, 3$. The closed itinerary

$$00 \longrightarrow 10 \longrightarrow 11 \longrightarrow 00$$

uses a two-bit-local 2-flip, a 3-flip, and a simultaneous flip. Assigning three distinct spectral parameters to these edge types and then changing the parameter-to-edge assignment changes the product in equation (14). The reproducibility certificate evaluates a fixed nonpalindromic witness, as well as a different endpoint path, at 80-digit precision; an independent checker recomputes both from the local blocks.

This assignment and path sensitivity is a genuine positive scope boundary: off-diagonal endpoint resolution retains data suppressed by the unprojected trace. It does not yet establish intrinsic chronology, because no source identifies the spectral parameter with a time step or defines the allowed event reorderings. It is not yet a positive Hilbert–Pólya candidate. A promotion would need all of the following.

1. A source-derived rule specifying which projected paths compose and which cyclic paths are primitive.
2. A repetition law compatible with the matrix weights and the chosen endpoint fibers.
3. A trace-class or nuclear operator whose flat trace yields those paths, not a finite product selected after the fact.
4. Meromorphic or entire continuation and a divisor different from the already known scattering and Kloosterman factors.

Kloosterman zeta functions and their twists already encode sophisticated off-diagonal arithmetic [13, 3], while relative scattering determinants are tightly related to Selberg zeta and resonances [1]. The projected witness is therefore recorded as a required next gate, not named as a new zeta.

Table 2: Object-wise Route-A ruling. “Fail” concerns the proposed Hilbert–Pólya route; “NT” means that the required object is not defined.

object	A1: primitive	A2: determinant	A3: divisor	A4: operator
rational cusp	fail: zero loop clock	fail: no periodic trace	fail: shifted quotient	NT
full boundary	pass: classical classes	NT: period set only	fail as new RH route	NT
bare products	NT: no sourced dynamics	NT: no orbit determinant	fail: shifted quotient	NT
projected paths	NT: no primitive law	NT: no Fredholm kernel	NT: no global divisor	NT

7 Route-A evaluation and reproducibility

The evaluation separates exact arithmetic from the stronger dynamical and operator claims required by Hilbert–Pólya.

For the rational-endpoint branch, the open arrows are geometric but the section-induced loop clock is zero and the open set has no canonical primitive-power monoid. The full-boundary enlargement passes A1 only by recovering classical hyperbolic conjugacy classes. We prove that its automorphy periods are signed Selberg lengths; we do not infer a determinant from that period set.

For bare squarefree scattering products, commutativity does not by itself forbid a finite or Fredholm determinant. The A1/A2 defects are instead that no source-derived evolution, primitive repetition law, or transfer kernel has been defined. Conditional on encoding steps by $\Phi_N(s_j)$, the frozen matrix product is permutation-invariant. The projector-resolved witness escapes that algebra but remains not testable at every Route-A gate beyond its finite assignment/path-sensitivity check.

A3: arithmetic divisor. The open coefficients and scattering matrices contain exact zeta arithmetic. Each standard eigenchannel still contains $\Lambda(2s-1)/\Lambda(2s)$, multiplied by local factors whose zeros and poles lie on $\Re s = 1$ and $\Re s = 0$. It therefore retains transported poles at $\rho/2$ and zeros at $(1+\rho)/2$, not one entire completed-zeta divisor.

Operator boundary. The hyperbolic Laplacian is self-adjoint and physical-line scattering is unitary. Those classical facts do not turn scattering resonances into a discrete point spectrum with ordinates of the Riemann zeros. The project defines no new Hilbert space, operator domain, or same-clock quantization.

7.1 Computational audit

The producer and checker implement three independent ledgers.

1. For every $q \leq 2000$, exact integer arithmetic computes $\varphi(q)$, all roots of -1 , and $n_q = (\varphi(q) + s_q)/2$. High-precision partial sums are compared with equation (3) in its half-plane of convergence.
2. Exact $\mathrm{SL}_2(\mathbb{Z})$ arithmetic checks the failed double-coset composition and the primitive/affine rational-section identities and their cocycle composition laws.
3. At levels 2, 6, 30, 210, direct Kronecker matrices are compared with Walsh reconstruction, functional equations, physical-line unitarity, determinants, spectral-parameter product

permutations, and the projected level-six witness. Three physical-line and two off-line points are used; calculations use 80 decimal digits.

The independent checker imports no producer code and recomputes the compact results at higher precision. The unit suite includes schema validation and tamper rejection. No Riemann-zero table, fitted spectral scale, or learned transition matrix is used.

The computational statements are regression certificates, not proofs of the universal theorems. All universal conclusions follow from the elementary arguments in the preceding sections.

8 Conclusion

Retaining modular endpoints resolves the compositional defect of cusp double cosets, but it changes the denominator height into an algebraically gauge-trivial cocycle on the rational orbit. Extending the endpoint space to all real points creates nonzero periods only by recovering signed hyperbolic translation lengths. Retaining all squarefree cusp channels also yields no order information in the frozen bare-matrix diagnostic: the standard scattering family belongs to one commutative Atkin–Lehner algebra and keeps the completed-zeta quotient divisor. This is not a claim that the spectral parameter is dynamical time.

These conclusions do not say that open scattering contains no dynamical information. The explicit open Laplace series is non-Selberg, and endpoint projectors restore assignment- and path-sensitive amplitudes. They show instead that the frozen ordinary diagonal closure discards the endpoint information that makes those objects open. Any successor must freeze the added off-diagonal state and derive its primitive law and operator before invoking a determinant.

The result is limited to projective automorphy cocycles and standard squarefree trivial-nebentypus scattering matrices. Nonsquarefree or twisted levels, matrix cocycles inserted between arrows, groupoid kernels with additional fiber state, and separately constructed self-adjoint operators remain outside the theorem. Because current Kloosterman and modular-symbol theory already occupies much of the nearest off-diagonal territory, the next breadth-first experiment should switch systems rather than tune the modular level. A natural next target is the two-axis Frobenius–iterate Hénon count, which preserves arithmetic extension degree and dynamical time separately.

A Supplementary identities and result schema

A.1 Local factor identities

Writing $M_p(s) = a_p(s)I + b_p(s)X$, direct algebra gives

$$a_p(s) + b_p(s) = \frac{1 + p^{1-s}}{1 + p^s}, \quad a_p(s) - b_p(s) = \frac{1 - p^{1-s}}{1 - p^s}.$$

Each ratio is inverted by $s \mapsto 1 - s$. Its numerator zeros have real part one and its denominator zeros have real part zero. These observations prove both the local functional equation and the noncancellation assertion in the open critical strips.

A.2 Projected trace identity

For $P_a = e_a e_a^T$ and arbitrary matrices A, B, C ,

$$P_a A P_b = A_{ab} e_a e_b^T.$$

Multiplying three such factors and taking the trace yields

$$\mathrm{tr}(P_a A P_b B P_c C) = A_{ab} B_{bc} C_{ca}.$$

This identity explains why projectors can retain an itinerary even when A, B, C commute.

A.3 Machine-readable outputs

The release stores the following compact artifacts.

- `arithmetic_counts.csv`: exact $q, \varphi(q), s_q, n_q$ rows through the frozen cutoff.
- `open_series.csv`: high-precision partial sums, closed forms, and elementary tail bounds.
- `endpoint_ledger.csv`: exact primitive and affine rational section checks.
- `exact_certificates.json`: symbolic identities, source locks, and exact representative witnesses.
- `scattering_checks.json`: Walsh channels, matrix identities, spectral-parameter product permutations, and projector-resolved controls.
- `summary.json`: frozen cutoffs, maximum residuals, and the scoped decision.
- `independent_check.json`: higher-precision independent verdict.

All complex numbers are serialized as decimal real/imaginary pairs. The checker rejects duplicate JSON keys, unexpected CSV columns, missing rows, and altered certificate values.

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